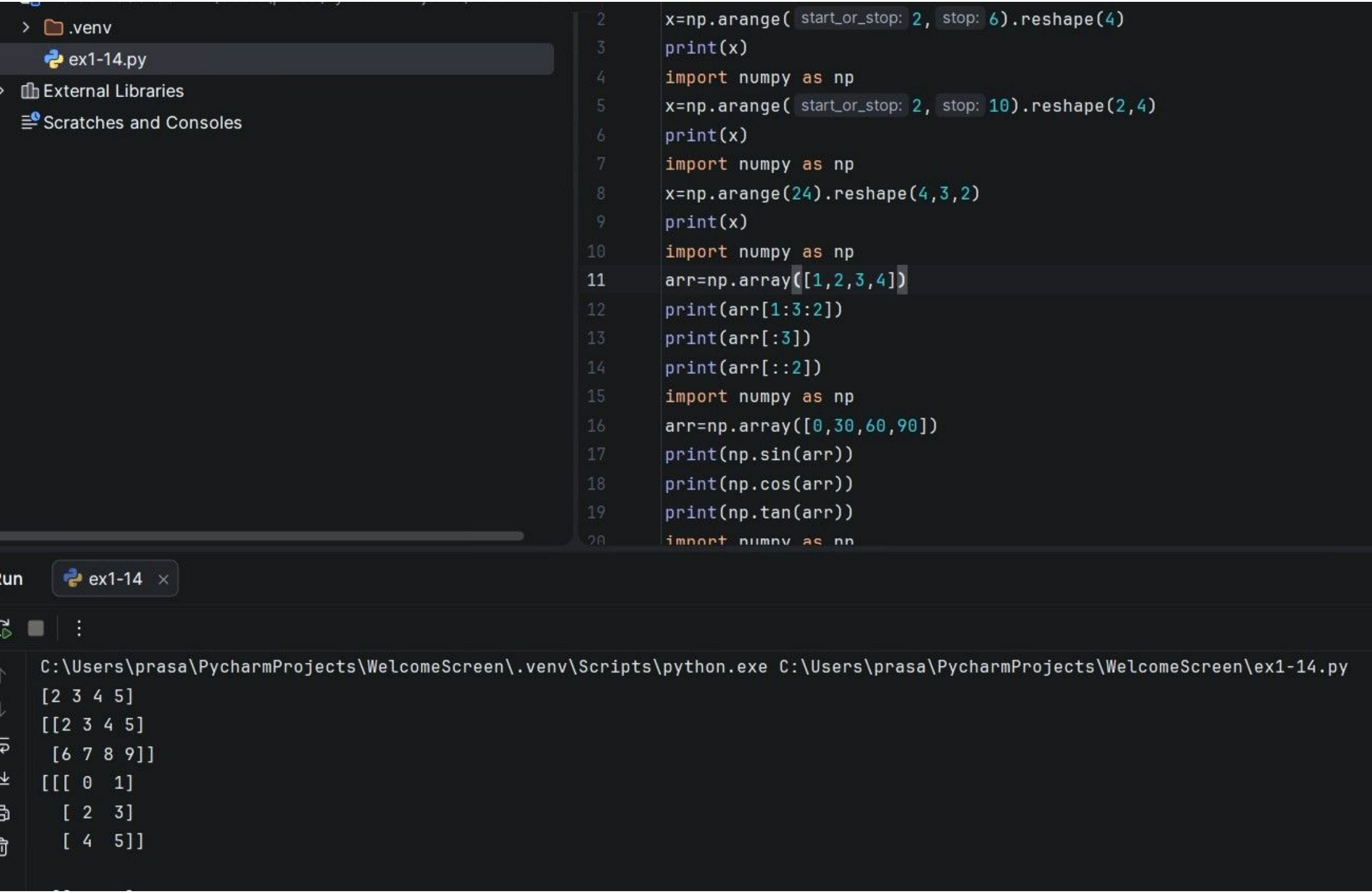

.venv
ex1-14.py
External Libraries
Scratches and Consoles
x=np.arange(start_or_stop: 2, stop: 6).reshape(4)
print(x)
import numpy as np
x=np.arange(start_or_stop: 2, stop: 10).reshape(2,4)
print(x)
import numpy as np
x=np.arange(24).reshape(4,3,2)
print(x)
import numpy as np
arr=np.array([1,2,3,4])
print(arr[1:3:2])
print(arr[:3])
print(arr[::2])
import numpy as np
arr=np.array([0,30,60,90])
print(np.sin(arr))
print(np.cos(arr))
print(np.tan(arr))
ex1-14
C:\Users\prasa\PycharmProjects\WelcomeScreen\.venv\Scripts\python.exe C:\Users\prasa\PycharmProjects\WelcomeScreen\ex1-14.py
[2 3 4 5]
[[2 3 4 5]
[6 7 8 9]]
[[[0 1]
[2 3]
[4 5]]

```
[2 3 4 5]
[[2 3 4 5]
 [6 7 8 9]]
[[[ 0  1]
  [ 2  3]
  [ 4  5]]

 [[ 6  7]
  [ 8  9]
  [10 11]]

 [[12 13]
  [14 15]
  [16 17]]

 [[18 19]
  [20 21]
  [22 23]]]
[2]
[1 2 3]
[1 3]
```

```
[ 0.         -0.98803182 -0.30481062  0.89599668]
[ 1.          0.15425145 -0.95241298 -0.44807362]
[ 0.         -6.4053312   0.32004039 -1.99520041]
Original List: [12.23, 13.32, 100, 36.32]
One-dimensional NumPy array:  [ 12.23  13.32 100.    36.32]
[12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
 36 37]
Original array:
[12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
 36 37]
Reverse array:
[37 36 35 34 33 32 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14
 13 12]
Original array:
[10, 20, 30]
After append values to the end of the array:
[10 20 30 40 50 60 70 80 90]
Array1:  [ 0 10 20 40 60]
Array2:  [10, 30, 40]
Common values between two arrays:
[10 40]
Array1:  [ 0 10 20 40 60 80]
Array2:  [10, 30, 40, 50, 70]
```

```
Array2:  [10, 30, 40, 50, 70]
Unique sorted array of values that are in either of the two input arrays:
[ 0 10 20 30 40 50 60 70 80]
[[ 0  1  3  0  2  4]
 [ 5  7  9  6  8 10]]
Values in Fahrenheit degrees:
[ 0.   12.   45.21 34.   99.91 32.  ]
Values in Centigrade degrees:
[-17.78 -11.11   7.34   1.11  37.73   0.  ]
Values in Centigrade degrees:
[-17.78 -11.11   7.34   1.11  37.73   0.  ]
Values in Fahrenheit degrees:
[-0.   12.   45.21 34.   99.91 32.  ]
First Array:
[[0. 1. 2.]
 [3. 4. 5.]
 [6. 7. 8.]]
Second Array:
[10 20 30]
Addition of Arrays:
[[10. 21. 32.]
 [13. 24. 35.]
 [16. 27. 38.]]
```

```
Subtraction of Arrays:
[[-10. -19. -28.]
 [ -7. -16. -25.]
 [ -4. -13. -22.]]
Multiplication of Arrays:
[[  0.  20.  60.]
 [ 30.  80. 150.]
 [ 60. 140. 240.]]
Division of Arrays:
[[0.         0.05       0.06666667]
 [0.3        0.2        0.16666667]
 [0.6        0.35       0.26666667]]
```

```
        name  score  attempts qualify
a  Anastasia   12.5         1     yes
b       Dima    9.0         3      no
c  Katherine   16.5         2     yes
d      James    NaN         3      no
e      Emily    9.0         2      no
f    Michael   20.0         3     yes
g    Matthew   14.5         1     yes
h      Laura    NaN         1      no
i      Kevin    8.0         2      no
j      Jonas   19.0         1     yes
Summary of the basic information about this DataFrame and its data:
<class 'pandas.DataFrame'>
Index: 10 entries, a to j
Data columns (total 4 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   name      10 non-null     str
 1   score     8 non-null      float64
 2   attempts  10 non-null     int64
 3   qualify   10 non-null     str
dtypes: float64(1), int64(1), str(2)
memory usage: 400.0+ bytes
None
```

```
C:\Users\prasa\PycharmProjects\PythonProject1\.venv\Scripts\python.exe C:\Users\prasa\PycharmProjects\PythonProject1\11259A504.py
1. NumPy Version:
2.5.4

2. One Dimensional Array:
[2 3 4 5]

3. Two Dimensional Array:
[[2 3 4 5]
 [6 7 8 9]]

4. Three Dimensional Array:
[[[ 0  1]
  [ 2  3]
  [ 4  5]]

 [[ 6  7]
  [ 8  9]
  [10 11]]

 [[12 13]
  [14 15]
  [16 17]]

 [[18 19]
  [20 21]
  [22 23]]]
```

```
5. Array Slicing:
[2]
[1 2 3]
[1 3]

6. Mathematical Functions:
Sin: [ 0.         -0.98803162 -0.30481062  0.89399666]
Cos: [ 1.          0.15425145 -0.95241298 -0.44807362]
Tan: [ 0.         -6.4053312   0.32004039 -1.99520041]

7. Convert List into Array:
Original List: [12.23, 13.32, 100, 36.32]
One-dimensional NumPy array: [ 12.23  13.32 100.    36.32]

8. Create Array:
[12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
 36 37]

9. Reverse Array:
Original array:
[12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
 36 37]
Reverse array:
[37 36 35 34 33 32 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14
 13 12]
```

```
10. Append Values:
Original array:
[10 20 30]
After append:
[10 20 30 40 50 60 70 80 90]

11. Common Values:
Array1: [ 0 10 20 40 60]
Array2: [10 30 40]
Common values:
[10 40]

12. Union of Arrays:
Array1: [ 0 10 20 40 60 80]
Array2: [10 30 40 50 70]
Unique sorted values:
[ 0 10 20 30 40 50 60 70 80]

13. Concatenate Arrays:
[[ 0  1  3  0  2  4]
 [ 5  7  9  6  8 10]]

14. Fahrenheit to Celsius:
Values in Fahrenheit:
[ 0.   12.   45.21 34.   99.91 32.  ]
Values in Celsius:
[-17.78 -11.11   7.34   1.11  37.73   0.  ]
```

```
15. Celsius to Fahrenheit:
Values in Celsius:
[-17.78 -11.11   7.34   1.11  37.73   0.  ]
Values in Fahrenheit:
[-0.   12.   45.21 34.   99.91 32.  ]
```

```
16. Arithmetic Operations:
First Array:
[[0. 1. 2.]
 [3. 4. 5.]
 [6. 7. 8.]]
Second Array:
[10 20 30]
Addition of Arrays:
[[10. 21. 32.]
 [13. 24. 35.]
 [16. 27. 38.]]
Subtraction of Arrays:
[[-10. -19. -28.]
 [ -7. -16. -25.]
 [ -4. -13. -22.]]
Multiplication of Arrays:
[[  0.  20.  60.]
 [ 30.  80. 150.]
 [ 60. 140. 240.]]
Division of Arrays:
[[0.         0.05       0.06666667]
 [0.3        0.2        0.16666667]
 [0.6        0.35       0.26666667]]

Process finished with exit code 0
```

```
        name  score  attempts qualify
a  Anastasia   12.5         1     yes
b       Dima    9.0         3      no
c  Katherine   16.5         2     yes
d      James    NaN         3      no
e      Emily    9.0         2      no
f    Michael   20.0         3     yes
g    Matthew   14.5         1     yes
h      Laura    NaN         1      no
i      Kevin    8.0         2      no
j      Jonas   19.0         1     yes
Summary of the basic information about this DataFrame and its data:
<class 'pandas.DataFrame'>
Index: 10 entries, a to j
Data columns (total 4 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   name      10 non-null     str
 1   score     8 non-null      float64
 2   attempts  10 non-null     int64
 3   qualify   10 non-null     str
dtypes: float64(1), int64(1), str(2)
memory usage: 400.0+ bytes
None
```

```
First three rows of the data frame:
        name  score  attempts qualify
a  Anastasia   12.5         1     yes
b       Dima    9.0         3      no
c  Katherine   16.5         2     yes
Select specific columns:
        name  score
a  Anastasia   12.5
b       Dima    9.0
c  Katherine   16.5
d      James    NaN
e      Emily    9.0
f    Michael   20.0
g    Matthew   14.5
h      Laura    NaN
i      Kevin    8.0
j      Jonas   19.0
Number of attempts in the examination is greater than 2:
      name  score  attempts qualify
b     Dima    9.0         3      no
d    James    NaN         3      no
f  Michael   20.0         3     yes
```

```
Number of Rows: 10
Number of Columns: 4
Rows where score is missing:
    name  score  attempts qualify
d  James    NaN         3      no
h  Laura    NaN         1      no
Original rows:
         name  score  attempts qualify
a  Anastasia   12.5         1     yes
b       Dima    9.0         3      no
c  Katherine   16.5         2     yes
d      James    NaN         3      no
e      Emily    9.0         2      no
f    Michael   20.0         3     yes
g    Matthew   14.5         1     yes
h      Laura    NaN         1      no
i      Kevin    8.0         2      no
j      Jonas   19.0         1     yes
```

```
Append a new row:
Print all records after insert a new record:
        name  score  attempts qualify
a  Anastasia   12.5         1     yes
b       Dima    9.0         3      no
c  Katherine   16.5         2     yes
d      James    NaN         3      no
e      Emily    9.0         2      no
f    Michael   20.0         3     yes
g    Matthew   14.5         1     yes
h      Laura    NaN         1      no
i      Kevin    8.0         2      no
j      Jonas   19.0         1     yes
k     Suresh   15.5         1     yes

Delete the new row and display the original rows:
        name  score  attempts qualify
a  Anastasia   12.5         1     yes
b       Dima    9.0         3      no
c  Katherine   16.5         2     yes
d      James    NaN         3      no
e      Emily    9.0         2      no
f    Michael   20.0         3     yes
g    Matthew   14.5         1     yes
h      Laura    NaN         1      no
i      Kevin    8.0         2      no
j      Jonas   19.0         1     yes
```

```
Orginal rows:
        name  score  attempts qualify
a  Anastasia   12.5         1     yes
b       Dima    9.0         3      no
c  Katherine   16.5         2     yes
d      James    NaN         3      no
e      Emily    9.0         2      no
f    Michael   20.0         3     yes
g    Matthew   14.5         1     yes
h      Laura    NaN         1      no
i      Kevin    8.0         2      no
j      Jonas   19.0         1     yes
Sort the data frame first by 'name' in descending order, then by 'score' in ascending order:
        name  score  attempts qualify
a  Anastasia   12.5         1     yes
b       Dima    9.0         3      no
c  Katherine   16.5         2     yes
d      James    NaN         3      no
e      Emily    9.0         2      no
f    Michael   20.0         3     yes
g    Matthew   14.5         1     yes
h      Laura    NaN         1      no
i      Kevin    8.0         2      no
j      Jonas   19.0         1     yes
```

```
Orginal rows:
        name  score  attempts qualify
a  Anastasia   12.5         1     yes
b       Dima    9.0         3      no
c  Katherine   16.5         2     yes
d      James    NaN         3      no
e      Emily    9.0         2      no
f    Michael   20.0         3     yes
g    Matthew   14.5         1     yes
h      Laura    NaN         1      no
i      Kevin    8.0         2      no
j      Jonas   19.0         1     yes
Sort the data frame first by 'name' in descending order, then by 'score' in ascending order:
        name  score  attempts qualify
a  Anastasia   12.5         1     yes
b       Dima    9.0         3      no
c  Katherine   16.5         2     yes
d      James    NaN         3      no
e      Emily    9.0         2      no
f    Michael   20.0         3     yes
g    Matthew   14.5         1     yes
h      Laura    NaN         1      no
i      Kevin    8.0         2      no
j      Jonas   19.0         1     yes
```

```
Original rows:
        name  score  attempts qualify
a  Anastasia   12.5         1     yes
b       Dima    9.0         3      no
c  Katherine   16.5         2     yes
d      James    NaN         3      no
e      Emily    9.0         2      no
f    Michael   20.0         3     yes
g    Matthew   14.5         1     yes
h      Laura    NaN         1      no
i      Kevin    8.0         2      no
j      Jonas   19.0         1     yes

Change the name 'James' to ‘Shyam’:
        name  score  attempts qualify
a  Anastasia   12.5         1     yes
b       Dima    9.0         3      no
c  Katherine   16.5         2     yes
d      Shyam    NaN         3      no
e      Emily    9.0         2      no
f    Michael   20.0         3     yes
g    Matthew   14.5         1     yes
h      Laura    NaN         1      no
i      Kevin    8.0         2      no
j      Jonas   19.0         1     yes

Process finished with exit code 0
```

```
Highest Mark: 92
Up the Stack Trace Ctrl+Alt+Up
8571429
Maximum Temperature: 35
Minimum Temperature: 29
Average Temperature: 32.0
Total Sales: 23800
Average Sales: 3400.0
Even IDs: [102, 104, 106]
Odd IDs: [101, 103, 105]
Positive Values: 3
Negative Values: 3
Duplicate IDs: [101, 102]
Highest Price: 600
Lowest Price: 120
Updated Salaries: [27500.000000000004, 33000.0, 44000.0, 38500.0]
Attendance Percentage: 90.0
Passed Students Marks: [67, 89, 78, 55]
Total Bill: 550
quality : 1
service : 2
excellent : 1
good : 2
12 Child
25 Adult
45 Adult
67 Senior Citizen
15 Child
Electricity Bill: 750
```

```
Highest Expense: 18000
Month: 3

Process finished with exit code 0
```

```
excellent : 1
good : 2
quality : 1
service : 2
12 Child
25 Adult
45 Adult
67 Senior Citizen
15 Child
Electricity Bill: 750
Highest Expense: 18000
Month: 3
First 5 Records:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    str
dtypes: float64(4), int64(1), str(1)
memory usage: 7.2 KB
None

Missing Values:
Id               0
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64
```

First 5 Records:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

Dataset Information:

```
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    str
dtypes: float64(4), int64(1), str(1)
memory usage: 7.2 KB
None
```

```
Missing Values:
Id               0
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64

Encoded Species:
   SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  Species
0            5.1           3.5            1.4           0.2        0
1            4.9           3.0            1.4           0.2        0
2            4.7           3.2            1.3           0.2        0
3            4.6           3.1            1.5           0.2        0
4            5.0           3.6            1.4           0.2        0

Preprocessed Dataset:
   SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  Species
0      -0.900681      1.032057      -1.341272     -1.312977        0
1      -1.143017     -0.124958      -1.341272     -1.312977        0
2      -1.385353      0.337848      -1.398138     -1.312977        0
3      -1.506521      0.106445      -1.284407     -1.312977        0
4      -1.021849      1.263460      -1.341272     -1.312977        0
```

```
Dataset Shape: (150, 4)
Number of Samples: 150
Number of Features: 4
Target Classes: ['setosa' 'versicolor' 'virginica']

Dataset Splitting
-----------------
Training samples: 120
Testing samples : 30

Test Set Performance
--------------------
Test Accuracy: 0.9667

5-Fold Cross-Validation
-----------------------
Fold 1 Accuracy: 1.0
Fold 2 Accuracy: 1.0
Fold 3 Accuracy: 0.9333
Fold 4 Accuracy: 0.9667
Fold 5 Accuracy: 0.9667

Cross-Validation Results
------------------------
Average Accuracy: 0.9733
Standard Deviation: 0.0249

Process finished with exit code 0
```

```
Features:
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']

Dataset Shape: (150, 3)
Target Shape: (150,)

Training Samples: 120
Testing Samples : 30

Actual vs Predicted Values
--------------------------------
Actual: 4.7  Predicted: 4.13
Actual: 1.7  Predicted: 1.88
Actual: 6.9  Predicted: 7.03
Actual: 4.5  Predicted: 4.43
Actual: 4.8  Predicted: 4.93
Actual: 1.5  Predicted: 2.07
Actual: 3.6  Predicted: 3.85
Actual: 5.1  Predicted: 6.13
Actual: 4.5  Predicted: 5.02
Actual: 3.9  Predicted: 3.97

Model Evaluation
----------------------------
Mean Squared Error (MSE): 0.13
Root Mean Squared Error (RMSE): 0.3606
Mean Absolute Error (MAE): 0.2605
R2 Score: 0.9603
```

```
Model Coefficients
--------------------------------
Sepal Length : 0.7228
Sepal Width : -0.6358
Petal Width : 1.4675
Intercept: -0.2622

Process finished with exit code 0
```

```
Dataset Shape: (150, 4)
Number of Samples: 150
Number of Features: 4
Target Classes: ['setosa' 'versicolor' 'virginica']

Dataset Splitting
-----------------
Training samples: 120
Testing samples : 30

KNN Model Evaluation
--------------------
Accuracy : 0.9333
Precision: 0.9444
Recall   : 0.9333
F1-Score : 0.9327

Confusion Matrix
----------------
[[10  0  0]
 [ 0 10  0]
 [ 0  2  8]]

Process finished with exit code 0
```

```
Dataset Shape: (150, 4)
Number of Samples: 150
Number of Features: 4
Target Classes: ['setosa' 'versicolor' 'virginica']

Dataset Splitting
-----------------
Training samples: 120
Testing samples : 30

SVM Model Evaluation
--------------------
Accuracy : 1.0
Precision: 1.0
Recall   : 1.0
F1-Score : 1.0

Confusion Matrix
----------------
[[10  0  0]
 [ 0 10  0]
 [ 0  0 10]]

Process finished with exit code 0
```

```
Missing Values:
Id               0
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64

Encoded Species:
   SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  Species
0            5.1           3.5            1.4           0.2        0
1            4.9           3.0            1.4           0.2        0
2            4.7           3.2            1.3           0.2        0
3            4.6           3.1            1.5           0.2        0
4            5.0           3.6            1.4           0.2        0

Preprocessed Dataset:
   SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  Species
0      -0.900681      1.032057      -1.341272     -1.312977        0
1      -1.143017     -0.124958      -1.341272     -1.312977        0
2      -1.385353      0.337848      -1.398138     -1.312977        0
3      -1.506521      0.106445      -1.284407     -1.312977        0
4      -1.021849      1.263460      -1.341272     -1.312977        0
```

```
Dataset Shape: (150, 4)
Number of Samples: 150
Number of Features: 4
Target Classes: ['setosa' 'versicolor' 'virginica']

Dataset Splitting
-----------------
Training samples: 120
Testing samples : 30

Test Set Performance
--------------------
Test Accuracy: 0.9667

5-Fold Cross-Validation
-----------------------
Fold 1 Accuracy: 1.0
Fold 2 Accuracy: 1.0
Fold 3 Accuracy: 0.9333
Fold 4 Accuracy: 0.9667
Fold 5 Accuracy: 0.9667

Cross-Validation Results
------------------------
Average Accuracy: 0.9733
Standard Deviation: 0.0249

Process finished with exit code 0
```

```
Features:
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']

Dataset Shape: (150, 3)
Target Shape: (150,)

Training Samples: 120
Testing Samples : 30

Actual vs Predicted Values
--------------------------------
Actual: 4.7  Predicted: 4.13
Actual: 1.7  Predicted: 1.88
Actual: 6.9  Predicted: 7.03
Actual: 4.5  Predicted: 4.43
Actual: 4.8  Predicted: 4.93
Actual: 1.5  Predicted: 2.07
Actual: 3.6  Predicted: 3.85
Actual: 5.1  Predicted: 6.13
Actual: 4.5  Predicted: 5.02
Actual: 3.9  Predicted: 3.97

Model Evaluation
----------------------------
Mean Squared Error (MSE): 0.13
Root Mean Squared Error (RMSE): 0.3606
Mean Absolute Error (MAE): 0.2605
R2 Score: 0.9603
```

```
Model Coefficients
--------------------------------
Sepal Length : 0.7228
Sepal Width : -0.6358
Petal Width : 1.4675
Intercept: -0.2622

Process finished with exit code 0
```

```
Dataset Shape: (150, 4)
Number of Samples: 150
Number of Features: 4
Target Classes: ['setosa' 'versicolor' 'virginica']

Dataset Splitting
-----------------
Training samples: 120
Testing samples : 30

KNN Model Evaluation
--------------------
Accuracy : 0.9333
Precision: 0.9444
Recall   : 0.9333
F1-Score : 0.9327

Confusion Matrix
----------------
[[10  0  0]
 [ 0 10  0]
 [ 0  2  8]]

Process finished with exit code 0
```

```
Dataset Shape: (150, 4)
Number of Samples: 150
Number of Features: 4
Target Classes: ['setosa' 'versicolor' 'virginica']

Dataset Splitting
-----------------
Training samples: 120
Testing samples : 30

SVM Model Evaluation
--------------------
Accuracy : 1.0
Precision: 1.0
Recall   : 1.0
F1-Score : 1.0

Confusion Matrix
----------------
[[10  0  0]
 [ 0 10  0]
 [ 0  0 10]]

Process finished with exit code 0
```

```
Dataset Shape: (150, 4)
Number of Samples: 150
Number of Features: 4

Features:
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']

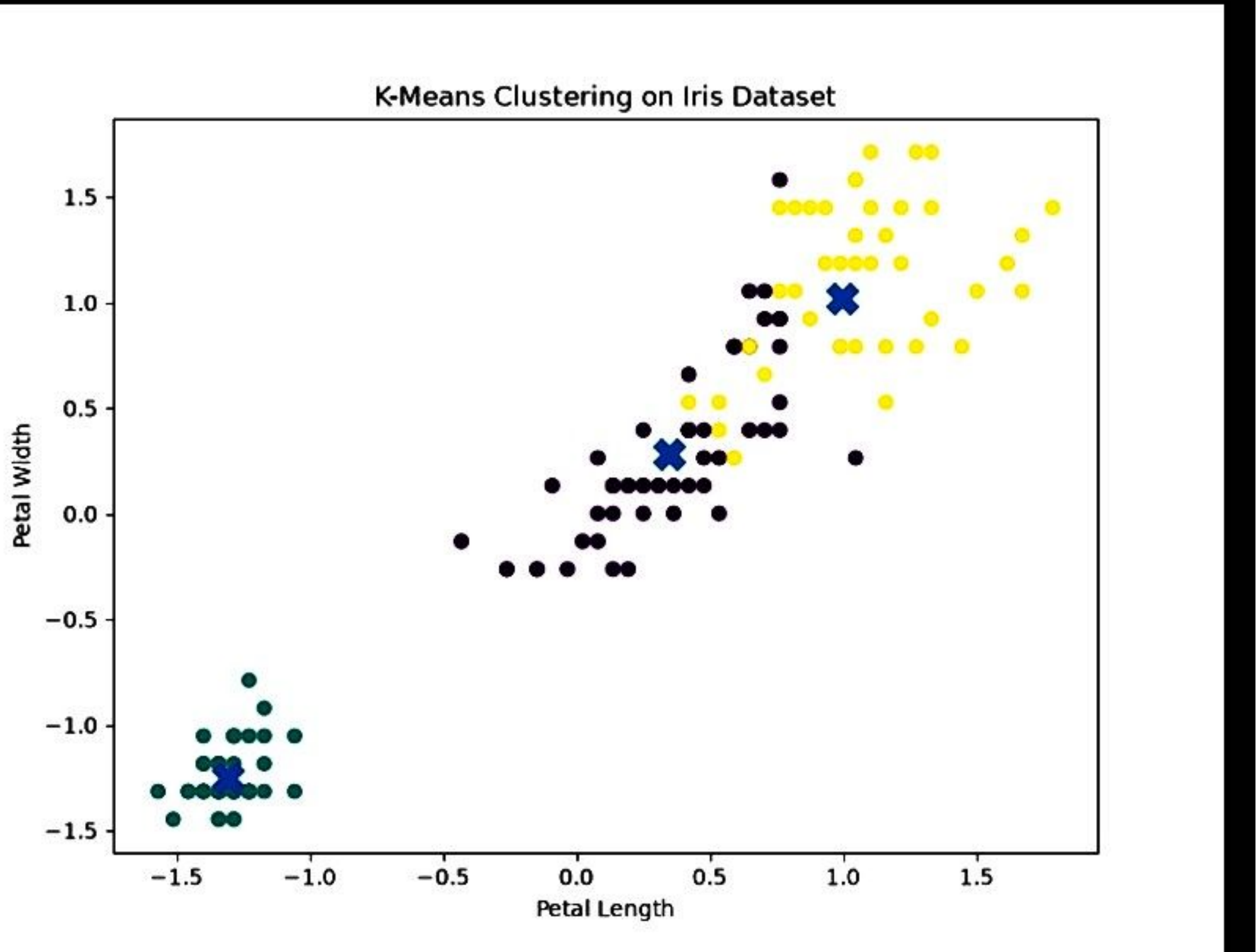
Cluster Labels:
[1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 0 0 0 2 0 0 0 0 0 0 0 2 0 0 0 0 2 0 0 0
 0 2 2 2 0 0 0 0 0 0 0 2 2 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0 2 2 2 2 0 2 2 2 2
 2 2 0 0 2 2 2 2 0 2 0 2 0 2 2 0 2 2 2 2 2 2 0 0 2 2 2 0 2 2 2 0 2 2 2 0 2
 2 0]

Cluster Centroids:
[[-0.05021989 -0.88337647  0.34773781  0.2815273 ]
 [-1.01457897  0.85326268 -1.30498732 -1.25489349]
 [ 1.13597027  0.08842168  0.99615451  1.01752612]]

Cluster Evaluation
------------------
Silhouette Score: 0.4599

Number of Samples in Each Cluster:
Cluster 0 : 53 samples
Cluster 1 : 50 samples
Cluster 2 : 47 samples

Process finished with exit code 0
```


K-Means Clustering on Iris Dataset
Petal Width
Petal Length
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5

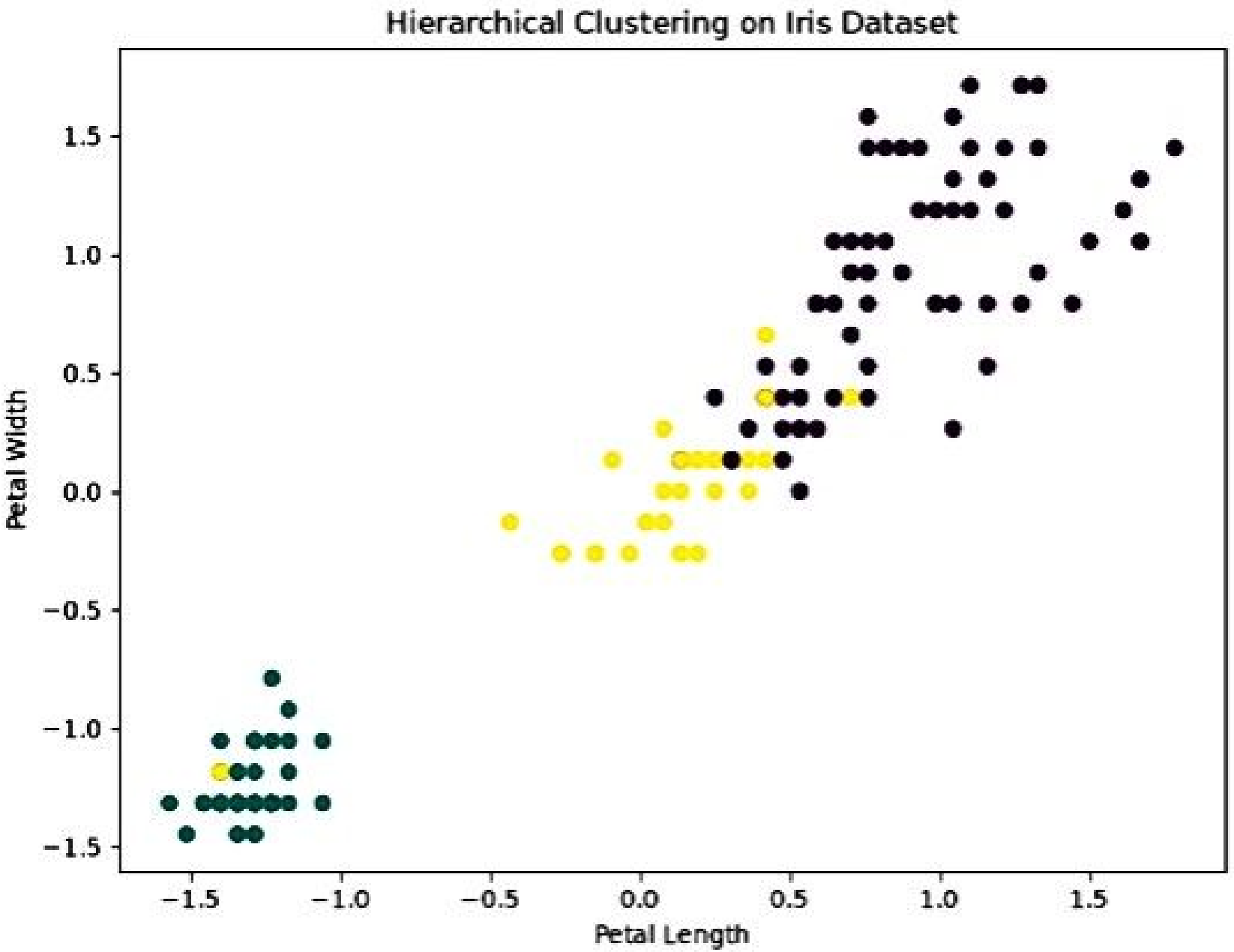
```
Dataset Shape: (150, 4)
Number of Samples: 150
Number of Features: 4

Features:
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']

Cluster Labels:
[1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 2 1 1 1 1 1 1 1 1 0 0 0 2 0 2 0 2 0 2 2 0 2 0 2 0 2 2 2 2 0 0 0 0
 0 0 0 0 0 2 2 2 2 0 2 0 0 2 2 2 2 0 2 2 2 2 2 0 2 2 0 0 0 0 0 0 2 0 0 0 0
 0 0 0 0 0 0 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0]

Silhouette Score: 0.4467

Number of Samples in Each Cluster:
Cluster 0 : 71 samples
Cluster 1 : 49 samples
Cluster 2 : 30 samples
```


Hierarchical Clustering on Iris Dataset
Petal Width
Petal Length
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5

```
Random Forest Classification
----------------------------
Number of trees: 100
Accuracy: 0.9

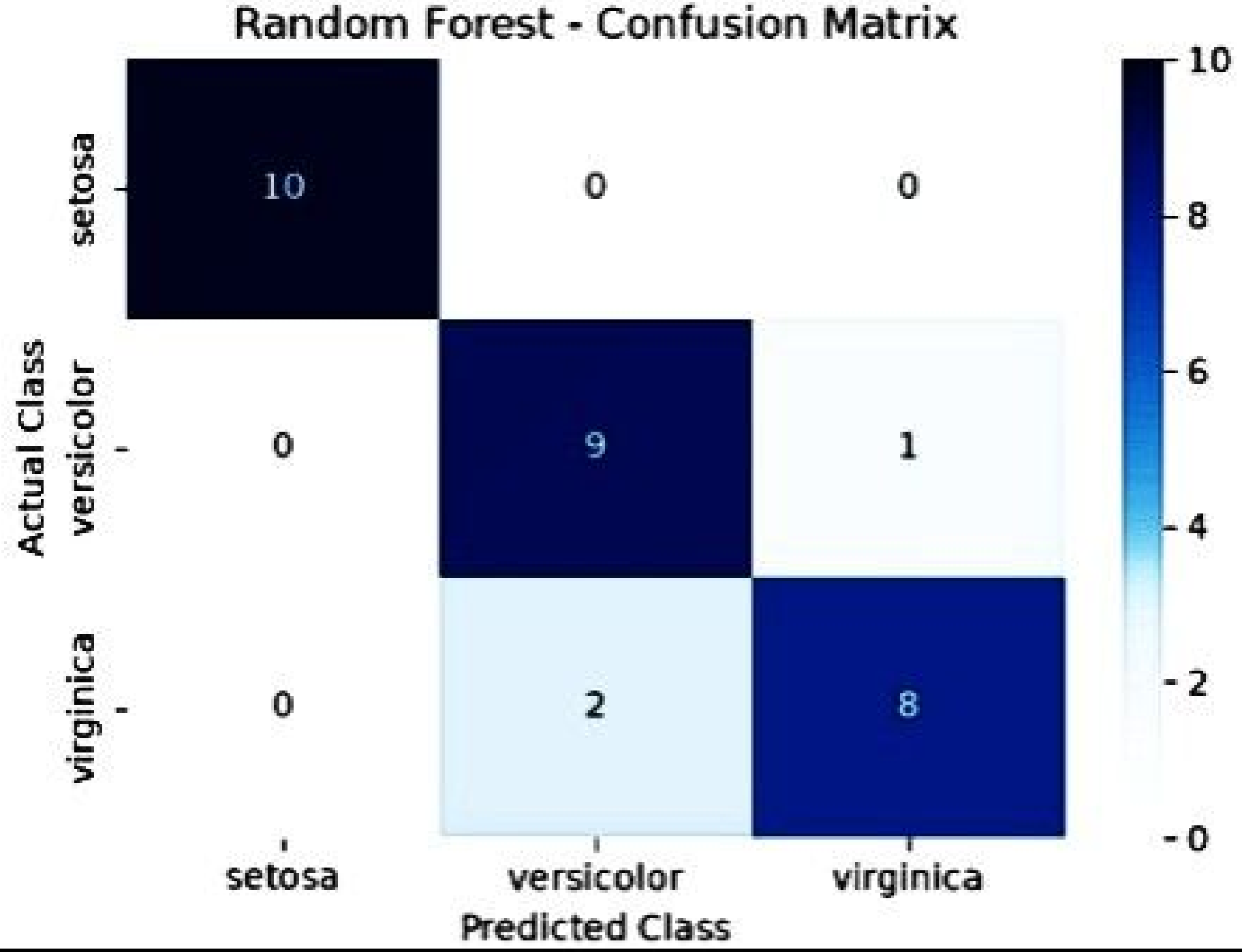
Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  2  8]]

Classification Report:
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       0.82      0.90      0.86        10
   virginica       0.89      0.80      0.84        10

    accuracy                           0.90        30
   macro avg       0.90      0.90      0.90        30
weighted avg       0.90      0.90      0.90        30


Process finished with exit code 0
```


Random Forest - Confusion Matrix
Actual Class
setosa
versicolor
virginica
10
0
0
0
9
1
0
2
8
setosa
versicolor
virginica
Predicted Class
10
8
6
4
2
0

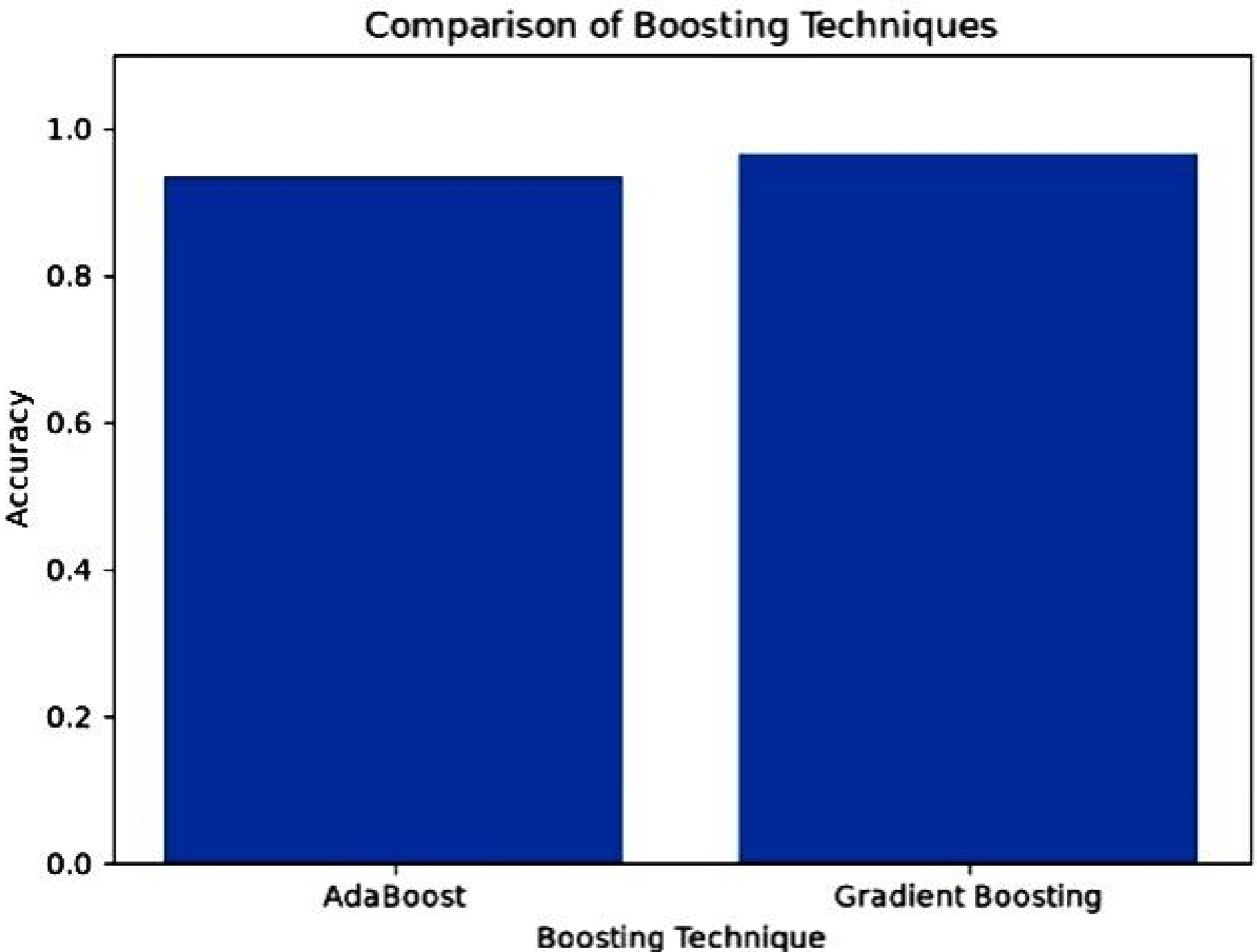
Classification Report:

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	0.82	0.90	0.86	10
virginica	0.89	0.80	0.84	10
accuracy			0.90	30
macro avg	0.90	0.90	0.90	30
weighted avg	0.90	0.90	0.90	30

```
 AdaBoost
------------------------------
Accuracy : 0.9333
Precision: 0.9333
Recall   : 0.9333
F1-Score : 0.9333

 Gradient Boosting
------------------------------
Accuracy : 0.9667
Precision: 0.9697
Recall   : 0.9667
F1-Score : 0.9666

Process finished with exit code 0
```


Comparison of Boosting Techniques
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
AdaBoost
Gradient Boosting
Boosting Technique

```
Learned Q-Table:
[[ 3.12191328  4.58        ]
 [ 3.12194542  6.2         ]
 [ 4.57999272  8.          ]
 [ 6.19995083 10.          ]
 [ 0.          0.        ]]

Learned Path:
[0, 1, 2, 3, 4]

Optimal Policy:
State 0 -> Right
State 1 -> Right
State 2 -> Right
State 3 -> Right

Process finished with exit code 0
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 16)	80
dense_1 (Dense)	(None, 8)	136
dense_2 (Dense)	(None, 3)	27

Total params: 243 (972.00 B)
Trainable params: 243 (972.00 B)
Non-trainable params: 0 (0.00 B)

```
Epoch 1/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 2s 23ms/step - accuracy: 0.4479 - loss: 1.1463 - val_accuracy: 0.6250 - val_loss: 1.0376
Epoch 2/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.4896 - loss: 1.0483 - val_accuracy: 0.6667 - val_loss: 0.9609
Epoch 3/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.6042 - loss: 0.9617 - val_accuracy: 0.7083 - val_loss: 0.8891
Epoch 4/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.6458 - loss: 0.8818 - val_accuracy: 0.7083 - val_loss: 0.8203
Epoch 5/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.7083 - loss: 0.8066 - val_accuracy: 0.7500 - val_loss: 0.7595
Epoch 6/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.7812 - loss: 0.7395 - val_accuracy: 0.7500 - val_loss: 0.7108
```

```
Epoch 7/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.8229 - loss: 0.6861 - val_accuracy: 0.7917 - val_loss: 0.6667
Epoch 8/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 1s 104ms/step - accuracy: 0.8542 - loss: 0.6383 - val_accuracy: 0.8333 - val_loss: 0.6317
Epoch 9/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 5s 194ms/step - accuracy: 0.8542 - loss: 0.5997 - val_accuracy: 0.8333 - val_loss: 0.6020
Epoch 10/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.8542 - loss: 0.5658 - val_accuracy: 0.8333 - val_loss: 0.5741
Epoch 11/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.8542 - loss: 0.5346 - val_accuracy: 0.8333 - val_loss: 0.5481
Epoch 12/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 2s 167ms/step - accuracy: 0.8542 - loss: 0.5069 - val_accuracy: 0.8333 - val_loss: 0.5240
Epoch 13/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.8542 - loss: 0.4804 - val_accuracy: 0.8333 - val_loss: 0.5026
Epoch 14/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - accuracy: 0.8542 - loss: 0.4565 - val_accuracy: 0.8333 - val_loss: 0.4824
Epoch 15/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 1s 38ms/step - accuracy: 0.8542 - loss: 0.4342 - val_accuracy: 0.8333 - val_loss: 0.4631
Epoch 16/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 1s 37ms/step - accuracy: 0.8646 - loss: 0.4126 - val_accuracy: 0.8333 - val_loss: 0.4435
Epoch 17/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 2s 119ms/step - accuracy: 0.8750 - loss: 0.3935 - val_accuracy: 0.8333 - val_loss: 0.4278
Epoch 18/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.8646 - loss: 0.3756 - val_accuracy: 0.8333 - val_loss: 0.4120
Epoch 19/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.8646 - loss: 0.3585 - val_accuracy: 0.8333 - val_loss: 0.3976
Epoch 20/50
```

```
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.8646 - loss: 0.3443 - val_accuracy: 0.8333 - val_loss: 0.3839
Epoch 21/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.8750 - loss: 0.3307 - val_accuracy: 0.8750 - val_loss: 0.3681
Epoch 22/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.8750 - loss: 0.3174 - val_accuracy: 0.8750 - val_loss: 0.3568
Epoch 23/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.8750 - loss: 0.3063 - val_accuracy: 0.8750 - val_loss: 0.3469
Epoch 24/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.8750 - loss: 0.2963 - val_accuracy: 0.8750 - val_loss: 0.3359
Epoch 25/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.8750 - loss: 0.2875 - val_accuracy: 0.8750 - val_loss: 0.3250
Epoch 26/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.8750 - loss: 0.2789 - val_accuracy: 0.8750 - val_loss: 0.3181
Epoch 27/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.8750 - loss: 0.2720 - val_accuracy: 0.8750 - val_loss: 0.3090
Epoch 28/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.8750 - loss: 0.2648 - val_accuracy: 0.8750 - val_loss: 0.2999
Epoch 29/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.8854 - loss: 0.2590 - val_accuracy: 0.8750 - val_loss: 0.2904
Epoch 30/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.8854 - loss: 0.2526 - val_accuracy: 0.8750 - val_loss: 0.2842
Epoch 31/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.8750 - loss: 0.2472 - val_accuracy: 0.8750 - val_loss: 0.2779
Epoch 32/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.8750 - loss: 0.2420 - val_accuracy: 0.8750 - val_loss: 0.2710
Epoch 33/50
```

```
Epoch 31/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.8750 - loss: 0.2472 - val_accuracy: 0.8750 - val_loss: 0.2779
Epoch 32/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.8750 - loss: 0.2420 - val_accuracy: 0.8750 - val_loss: 0.2710
Epoch 33/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.8958 - loss: 0.2383 - val_accuracy: 0.8750 - val_loss: 0.2576
Epoch 34/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9062 - loss: 0.2314 - val_accuracy: 0.8750 - val_loss: 0.2500
Epoch 35/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.9062 - loss: 0.2263 - val_accuracy: 0.8750 - val_loss: 0.2431
Epoch 36/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9062 - loss: 0.2213 - val_accuracy: 0.8750 - val_loss: 0.2366
Epoch 37/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9062 - loss: 0.2169 - val_accuracy: 0.8750 - val_loss: 0.2281
Epoch 38/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.9167 - loss: 0.2128 - val_accuracy: 0.8750 - val_loss: 0.2204
Epoch 39/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9167 - loss: 0.2076 - val_accuracy: 0.8750 - val_loss: 0.2150
Epoch 40/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9167 - loss: 0.2034 - val_accuracy: 0.8750 - val_loss: 0.2086
Epoch 41/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9271 - loss: 0.1994 - val_accuracy: 0.9167 - val_loss: 0.2032
Epoch 42/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9271 - loss: 0.1955 - val_accuracy: 0.9583 - val_loss: 0.1957
Epoch 43/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9167 - loss: 0.1922 - val_accuracy: 0.9583 - val_loss: 0.1879
```

```
Epoch 44/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9271 - loss: 0.1883 - val_accuracy: 0.9583 - val_loss: 0.1865
Epoch 45/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9167 - loss: 0.1852 - val_accuracy: 0.9583 - val_loss: 0.1780
Epoch 46/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9167 - loss: 0.1813 - val_accuracy: 0.9583 - val_loss: 0.1745
Epoch 47/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.9167 - loss: 0.1778 - val_accuracy: 0.9583 - val_loss: 0.1710
Epoch 48/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9167 - loss: 0.1740 - val_accuracy: 0.9583 - val_loss: 0.1647
Epoch 49/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9271 - loss: 0.1720 - val_accuracy: 0.9583 - val_loss: 0.1623
Epoch 50/50
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9271 - loss: 0.1688 - val_accuracy: 0.9583 - val_loss: 0.1534

Test Loss: 0.2352
Test Accuracy: 0.8667

Classification Report:
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       0.88      0.70      0.78        10
   virginica       0.75      0.90      0.82        10
```

```
    accuracy                           0.87        30
   macro avg       0.88      0.87      0.87        30
weighted avg       0.88      0.87      0.87        30


Process finished with exit code 0
```